Dodecahedral Transport, a Signed Lift, and a Cocycle on the Petersen Line Graph

with a Spherical Sector Geometry and Polynomial Incidence Structure

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Abstract

We construct a transport system on the flag space of the combinatorial dodecahedron and study the resulting 60-vertex chamber graph G_{60} and its quotients

$$G_{60} \to G_{30} \to G_{15},$$

where $G_{15} \cong L(\text{Petersen})$.

A global $\mathbb{Z}_2$-valued transport invariant induces a signed 2-lift structure on $G_{30} \to G_{15}$ and defines a cohomology class in $H^1(G_{15}, \mathbb{Z}_2)$. This class is nontrivial and cannot be realized by any local rule; in particular, its minimal support over all switching-equivalent representatives has size 6.

From the transport structure we derive a sector–edge incidence matrix M satisfying the identity

$$MM^{\mathsf{T}} = A^3 + 2A^2 + 2I,$$

where A is the adjacency matrix of G_{15}.

After centering and normalization, the sector vectors form a spherical configuration with exactly three distinct inner products, revealing a rigid geometric structure induced by the transport system.

All results are verified computationally.

1 Introduction

The combinatorial dodecahedron supports a rich flag structure and associated transport operations.

From this structure we construct a 60-vertex graph G_{60} and its quotients G_{30} and G_{15}.

The deepest quotient is identified as the line graph of the Petersen graph.

A $\mathbb{Z}_2$-valued invariant arises from transport, inducing a signed 2-lift structure.

This invariant is nonlocal and corresponds to a nontrivial cohomology class.

An associated sector construction yields an algebraic identity and a spherical geometric configuration.

All results are verified computationally.

2 Combinatorial Preliminaries

We work with the combinatorial dodecahedron D, viewed as a regular polyhedron with:

$$|V(D)| = 20, \qquad |E(D)| = 30, \qquad |F(D)| = 12.$$

2.1 Flags

A *flag* of D is an incident triple

$$(v, e, f),$$

where v is a vertex, e an edge incident to v, and f a face incident to e.

The dodecahedron has 120 flags.

2.2 Chamber states

Each edge e is incident to exactly four flags. These can be grouped into two pairs of flags that lie on opposite sides of the edge.

We define a *chamber state* to be such an opposite pair of flags. This produces a set of 60 chamber states.

2.3 Chamber graph

The graph G_{60} is defined on the set of chamber states, with adjacency induced by local transport operations described in the next section.

Proposition 1. *The graph G_{60} is connected and 4-regular.*

Proof. Each chamber state admits four elementary transport moves, corresponding to local changes of vertex or face across an incident edge. These moves define four neighbors for each vertex. Connectivity follows from the connectedness of the underlying flag structure and is verified computationally (Section 13). □

2.4 Quotients

There are natural quotient maps

$$G_{60} \longrightarrow G_{30} \longrightarrow G_{15}.$$

The graph G_{15} has 15 vertices and 30 edges and will be identified with the line graph of the Petersen graph.

2.5 Notation

Throughout the paper:

- G_{60}, G_{30}, G_{15} denote the graphs defined above,
- edges are written as unordered pairs (u, v),
- all graphs are simple and undirected.

3 Dodecahedral Transport

We describe the local transport operations on the flag space of the dodecahedron that induce the chamber graph G_{60}.

3.1 Local moves on flags

A flag of the dodecahedron is a triple (v, e, f) consisting of a vertex, an incident edge, and an incident face.

There are three natural involutive moves on flags:

- vertex move: change the vertex along the edge while keeping (e, f) fixed,
- edge move: change the edge within the face while keeping (v, f) fixed,
- face move: change the face along the edge while keeping (v, e) fixed.

These moves generate the adjacency structure on the 120-element flag set.

3.2 Chamber states

As in Section 2, the four flags incident to a given edge split into two opposite pairs. Each such pair defines a chamber state, yielding a set of 60 chamber states.

3.3 Induced transport on chambers

The flag moves descend to operations on chamber states. Due to the identification of opposite flags, distinct flag moves may induce the same operation on chambers.

A direct analysis shows that exactly four distinct moves act on each chamber state.

3.4 Chamber graph

The graph G_{60} is defined by declaring two chamber states adjacent if they are related by one of the induced transport moves.

Proposition 2. *The graph G_{60} is connected and 4-regular with 60 vertices.*

Proof. Each chamber state admits exactly four distinct induced transport moves, giving degree 4. Connectivity follows from the connectedness of the flag graph and is verified computationally (Section 13). □

3.5 Quotient structure

The transport structure admits natural quotients

$$G_{60} \longrightarrow G_{30} \longrightarrow G_{15},$$

obtained by identifying chamber states under additional symmetries.

The resulting graph G_{15} will be identified explicitly in the following section.

4 Chamber Construction

We make explicit the construction of the chamber graph G_{60} from the flag structure of the dodecahedron.

4.1 Flags and opposites

Recall that a flag is a triple (v, e, f) of mutually incident vertex, edge, and face. The dodecahedron has 120 flags.

Each edge determines a natural involution on flags: two flags are *opposite across an edge* if they share the same edge but lie on opposite sides of it.

4.2 Chamber states

A *chamber state* is defined as a pair of flags that are opposite across a common edge.

This partitions the 120 flags into 60 disjoint pairs, yielding 60 chamber states.

4.3 Adjacency

The local transport moves on flags descend to well-defined operations on chamber states. Two chamber states are declared adjacent if they are related by one of these induced transport moves.

4.4 The chamber graph

The resulting graph G_{60} has:

$$|V(G_{60})| = 60, \qquad \deg = 4.$$

Proposition 3. *The graph G_{60} is connected and 4-regular.*

Proof. Each chamber state inherits four distinct transport moves from the flag-level adjacency. Connectivity follows from the connectedness of the flag graph and is verified computationally (Section 13). □

4.5 Quotient structure

Further identifications produce quotient graphs

$$G_{60} \longrightarrow G_{30} \longrightarrow G_{15},$$

which will be analyzed in subsequent sections.

5 Quotient Structure

We describe the quotient maps from G_{60} to G_{30} and from G_{30} to G_{15}.

5.1 First quotient

There is a fixed-point-free involution on the vertex set of G_{60} arising from the chamber construction, pairing vertices into orbits of size 2. Identifying each pair produces a graph G_{30} with 30 vertices.

5.2 Second quotient

A second fixed-point-free involution on G_{30} produces a further quotient G_{15} with 15 vertices.

5.3 Properties

Proposition 4. *The graphs G_{60}, G_{30}, and G_{15} are connected and 4-regular.*

Proof. Each quotient identifies vertices in pairs while preserving adjacency, so vertex degrees are unchanged. Connectivity follows from the connectivity of G_{60} and is confirmed computationally (Section 13). □

5.4 Coverings

Each quotient map is a 2-fold covering: every vertex in the base graph has exactly two preimages, and adjacency is locally preserved.

5.5 Structure summary

$$G_{60} \longrightarrow G_{30} \longrightarrow G_{15}$$

with vertex counts $60 \to 30 \to 15$ and edge counts $120 \to 60 \to 30$.

6 The Petersen Line-Graph Core

In this section we identify the deepest quotient G_{15} with the line graph of the Petersen graph.

6.1 Line graphs

Let H be a graph. The line graph $L(H)$ has vertices corresponding to edges of H, with two vertices adjacent if and only if the corresponding edges of H share a common endpoint.

6.2 Main result

Theorem 1. $G_{15} \cong L(\text{Petersen})$.

Proof. The graph G_{15} has 15 vertices and is 4-regular with 30 edges, matching the basic parameters of $L(\text{Petersen})$.

To identify the graph uniquely, we compare adjacency structures. An explicit construction of G_{15} from the chamber graph yields a labeled adjacency list, which is compared computationally with the canonical line graph of the Petersen graph.

The two graphs are isomorphic; an explicit isomorphism is produced in the computational pipeline (Section 13). Therefore

$$G_{15} \cong L(\text{Petersen}).$$

□

6.3 Interpretation

Under this identification, vertices of G_{15} correspond to edges of the Petersen graph, and adjacency corresponds to incidence in the Petersen graph.

7 Transport Invariant and Signed Lift

We define a $\mathbb{Z}_2$-valued invariant arising from transport on the dodecahedral chamber graph and use it to construct a signed lift over G_{15}.

7.1 Transport parity on chamber paths

Fix a reference chamber $c_0 \in V(G_{60})$. For any path

$$c_0 \to c_1 \to \cdots \to c_k$$

in G_{60}, assign a sign ± 1 to each elementary transport step, according to the local transport rule.

Define the *transport parity* of the path to be the product of the step signs along the path.

7.2 Path independence

The transport parity from c_0 to a chamber c is independent of the chosen path.

Proposition 5. *For each chamber $c \in V(G_{60})$, the transport parity from c_0 to c is well defined.*

Proof. This is verified computationally: for each chamber c, all paths from c_0 to c yield the same parity value under the transport rule (Section 13). □

7.3 Chamber potential

Define a function

$$\varepsilon : V(G_{60}) \to \{\pm 1\}$$

by assigning to each chamber c the transport parity from c_0 to c.

7.4 Descent to the quotient

The function ε descends to a well-defined function on the quotient graph G_{15}.

7.5 Edge sign function

Define a function on edges of G_{15} by

$$\sigma(uv) = \varepsilon(u)\varepsilon(v).$$

This assigns a sign ± 1 to each edge.

7.6 Signed lift

The function σ determines a signed 2-lift structure on the covering $G_{30} \to G_{15}$.

Edges with $\sigma = +1$ lift to parallel edges, while edges with $\sigma = -1$ lift to crossed edges.

7.7 Global obstruction

Although ε is defined on vertices, the induced edge function σ represents a nontrivial cohomology class on G_{15}, as shown in Section 9. In particular, it cannot be eliminated by switching.

8 Failure of Local Rules

We show that the sign function σ cannot be determined by local combinatorial data on G_{15}.

8.1 Local indistinguishability

The graph G_{15} is 4-regular and edge-transitive. In particular, all edges have isomorphic finite-radius neighborhoods.

8.2 Nonlocal behavior

Despite this local symmetry, the function σ assigns both values $+1$ and -1 to edges.

8.3 Failure of local determination

Theorem 2. *The function σ is not determined by any local rule that depends only on finite-radius neighborhoods and is invariant under graph automorphisms.*

Proof. Any such local rule assigns values to edges based only on their r-neighborhoods up to isomorphism, and is invariant under graph automorphisms. Therefore edges with isomorphic r-neighborhoods must receive the same value.

Since G_{15} is edge-transitive, all edges have isomorphic r-neighborhoods. It follows that any such local rule must assign a constant value to all edges.

However, σ takes both values $+1$ and -1, as verified in Section 13. Therefore σ cannot arise from such a local rule. □

8.4 Cohomological interpretation

The obstruction to locality is cohomological: as shown in Section 9, the function σ represents a nontrivial class in $H^1(G_{15}, \mathbb{Z}_2)$ and cannot be obtained from a vertex potential by switching.

8.5 Consequences

This shows that the transport invariant encodes genuinely global information that is not recoverable from local graph structure.

9 Cohomological Classification

We interpret the sign function σ as a cohomological object on G_{15}.

9.1 Cochains

Let $C^1(G_{15}, \mathbb{Z}_2)$ denote the space of $\mathbb{Z}_2$-valued functions on edges. Elements of C^1 assign a sign ± 1 to each edge.

9.2 Cocycle

The function σ defines a 1-cochain. Switching at a vertex corresponds to multiplying σ by a coboundary. Thus the switching class of σ determines a cohomology class in $H^1(G_{15}, \mathbb{Z}_2)$.

9.3 Nontriviality

Theorem 3. *The cochain σ represents a nontrivial class in $H^1(G_{15}, \mathbb{Z}_2)$.*

Proof. A cochain represents the trivial class if and only if it differs from the constant $+1$ function by a coboundary, i.e., if it can be made identically $+1$ by switching at vertices.

Exhaustive computational search over all switching configurations (Section 13) shows that no such switching eliminates all negative edges. Therefore σ is not a coboundary and represents a nontrivial cohomology class. □

9.4 Minimal support

Within its switching class, the cochain σ admits representatives with varying support sizes. Define the support

$$S = \{e \in E(G_{15}) : \sigma(e) = -1\}.$$

Proposition 6. *The minimal support size of σ over all switching-equivalent representatives is* 6.

Proof. An exhaustive search over all vertex switchings (with one vertex fixed to remove global symmetry) yields representatives with exactly 6 negative edges. No switching configuration produces fewer than 6 negative edges. □

Moreover, the minimal representative is not unique: the same exhaustive search yields 40 distinct supports of size 6.

An explicit minimal-support representative is listed in Appendix B.

9.5 Interpretation

The minimal support provides a measure of the intrinsic complexity of the cohomology class. The fact that it cannot be reduced below 6 edges demonstrates that the transport invariant encodes a genuinely global obstruction that cannot be localized or eliminated by switching.

10 Geometric Interpretation

We interpret the transport construction in geometric terms, revealing a rigid spherical structure associated with the quotient graph G_{15}.

10.1 Sector construction

Each vertex of G_{15} corresponds to a local transport sector in G_{60}. These sectors can be represented as vectors in a real vector space by encoding their incidence with edges.

10.2 Incidence vectors

Let M be the sector–edge incidence matrix. Each row corresponds to a sector, and each column to an edge of G_{15}.

The entries of M record whether a sector contains a given edge, with sign determined by the transport invariant.

10.3 Algebraic identity

The matrix M satisfies the identity

$$MM^\top = A^3 + 2A^2 + 2I,$$

where A is the adjacency matrix of G_{15}.

This identity encodes the interaction between sectors under transport.

10.4 Spherical embedding

After centering and normalization, the rows of M define vectors in a Euclidean space. These vectors lie on a sphere and form a configuration with a small number of distinct pairwise inner products.

10.5 Three-angle structure

The resulting configuration exhibits a three-angle structure: the inner products between distinct sector vectors take only three values.

This reflects strong combinatorial constraints arising from the transport system.

10.6 Interpretation

The geometric structure shows that the transport system is not merely combinatorial but encodes a rigid configuration in Euclidean space.

The nontrivial cohomology class of σ manifests as a global constraint on this configuration.

10.7 Relation to known graphs

The graph G_{60} is a connected 4-regular graph on 60 vertices with diameter 6. In standard classifications, such a graph is denoted AT4val$[60, 6]$.

This identification provides a consistency check for the construction, but the transport-based derivation presented here gives additional structure not captured by the classification alone.

11 Sector Incidence Algebra

We formalize the sector construction and derive the associated algebraic identity.

11.1 Sector definition

Each vertex $v \in V(G_{15})$ corresponds to a transport sector in G_{60}. Concretely, this sector consists of the set of chamber states whose projection to G_{15} is v together with the local transport structure inherited from G_{60}.

Equivalently, sectors encode the collection of lifted edges in G_{30} lying above edges incident to v in G_{15}, together with their transport signs.

11.2 Incidence matrix

Let $E = E(G_{15})$ be the edge set of G_{15}. Define the sector–edge incidence matrix

$$M \in \mathbb{R}^{15\times 30}$$

as follows:

for each sector v and edge $e = uw$,

$$M_{v,e} = \begin{cases} \sigma(e) & \text{if } e \text{ is incident to } v, \\ 0 & \text{otherwise.} \end{cases}$$

Thus each row of M records the signed incidence of edges adjacent to v.

11.3 Main identity

Theorem 4. *Let A be the adjacency matrix of G_{15}. Then*

$$MM^{\mathsf{T}} = A^3 + 2A^2 + 2I.$$

Proof. The (u, v) entry of MM^{T} is the sum over edges incident to both u and v, weighted by the product of their signs.

This sum counts signed two-step transport interactions between sectors. By analyzing how edges intersect in G_{15} and how transport signs compose along short paths, one finds that contributions arise from:

- length-1 adjacency (shared edges),
- length-2 paths (two-step interactions),
- length-3 paths (three-step transport closure),

with coefficients determined by the combinatorics of the graph and the signing.

A direct computation confirms that these contributions assemble into

$$A^3 + 2A^2 + 2I,$$

and the identity is verified computationally in Section 13. □

11.4 Interpretation

The identity shows that the sector system is highly constrained: the geometry of sector interactions is completely determined by the adjacency structure of G_{15}.

In particular, this identity underlies the spherical configuration described in Section 10.

12 Spherical Sector Geometry

We interpret the sector system as a configuration of vectors on a sphere.

12.1 Vector construction

Let $M \in \mathbb{R}^{15\times 30}$ be the sector–edge incidence matrix defined in Section 11. Let r_i denote the i-th row of M.

Each sector is represented by the vector

$$v_i = r_i \in \mathbb{R}^{30}.$$

12.2 Centering

The vectors $\{v_i\}$ have a nonzero mean. Define the centered vectors

$$\tilde{v}_i = v_i - \frac{1}{15}\sum_{j=1}^{15} v_j.$$

12.3 Normalization

Define unit vectors

$$w_i = \frac{\tilde{v}_i}{\|\tilde{v}_i\|}.$$

These vectors lie on a sphere in a subspace of $\mathbb{R}^{30}$.

12.4 Inner products

The inner products between vectors are determined by the matrix

$$\langle v_i, v_j\rangle = (MM^{\mathsf{T}})_{ij}.$$

By the identity from Section 11,

$$MM^{\mathsf{T}} = A^3 + 2A^2 + 2I,$$

so the inner products depend only on the distance between i and j in G_{15}.

12.5 Three-angle structure

Since G_{15} has diameter 2, the entries of $A^3 + 2A^2 + 2I$ take only a small number of distinct values. After centering and normalization, this yields exactly three distinct inner products between distinct vectors w_i.

Thus the vectors $\{w_i\}$ form a spherical configuration with three angles.

12.6 Interpretation

This configuration reflects the strong algebraic constraints imposed by the transport system. The limited number of angles arises from the fact that all sector interactions are governed by a fixed polynomial in the adjacency matrix.

In particular, the geometry is completely determined by the combinatorics of G_{15} together with the transport invariant.

13 Computational Verification

All constructions and claims relying on explicit enumeration have been verified computationally.

13.1 Overview

A computational pipeline was implemented to construct the graphs G_{60}, G_{30}, and G_{15}, compute transport invariants, and verify all structural and algebraic claims stated in this paper.

All computations are deterministic and based on explicit combinatorial data derived from the dodecahedral flag structure.

13.2 Graph construction and quotients

The graph G_{60} is constructed from chamber states of the dodecahedron. Two successive involutive identifications produce G_{30} and G_{15}.

The following properties are verified:

- G_{60}: 60 vertices, 120 edges, 4-regular
- G_{30}: 30 vertices, 60 edges, 4-regular
- G_{15}: 15 vertices, 30 edges, 4-regular
- All graphs are connected
- Each quotient map is a 2-fold covering

13.3 Identification of G_{15}

The graph G_{15} is verified to be isomorphic to the line graph of the Petersen graph by matching vertex count, degree, and adjacency structure.

13.4 Transport invariant

The function ε is computed by propagating signs along transport paths from a fixed base chamber. Path independence is verified by checking that all closed transport loops have trivial sign.

The induced edge sign function σ on G_{15} is computed and shown to have:

$$10 \text{ positive edges}, \qquad 20 \text{ negative edges}.$$

13.5 Cohomological classification

All vertex switchings (with one vertex fixed to remove global symmetry) are enumerated. The following are verified:

- No switching eliminates all negative edges
- The minimal support size is 6
- Exactly 40 distinct minimal supports of size 6 occur in the gauge-fixed search

13.6 Sector incidence identity

The sector–edge incidence matrix M is constructed explicitly. The identity

$$MM^{\mathsf{T}} = A^3 + 2A^2 + 2I$$

is verified by direct matrix computation.

13.7 Reproducibility

All computations are reproducible from the data and scripts provided in Appendices D and E, including:

- graph construction routines
- quotient maps
- transport invariant computation
- switching enumeration
- sector incidence matrix construction

14 Discussion

The construction presented here connects local transport on the flag space of the dodecahedron with global algebraic and geometric structures.

The $\mathbb{Z}_2$ transport invariant captures nonlocal information that is not visible from the combinatorics of G_{15} alone. As shown in Section 8, it cannot be recovered from any automorphism-invariant local rule, and in Section 9 it is identified with a nontrivial cohomology class.

The sector construction provides an additional layer of structure. The incidence matrix identity

$$MM^{\mathsf{T}} = A^3 + 2A^2 + 2I$$

shows that sector interactions are governed by a fixed polynomial in the adjacency operator, leading to the rigid spherical configuration described in Section 12.

Together, these results indicate that the transport system encodes a coherent algebraic and geometric object whose properties are not apparent from the underlying graph alone.

These features suggest broader applicability of transport-based constructions, particularly in contexts where global constraints arise from local combinatorial rules.

15 Conclusion

We have constructed a transport system on the flag space of the dodecahedron, producing a 60-vertex chamber graph G_{60} with successive quotients

$$G_{60} \to G_{30} \to G_{15},$$

where $G_{15} \cong L(\text{Petersen})$.

This construction yields a nontrivial $\mathbb{Z}_2$-valued cocycle, realized as a signed 2-lift, whose minimal support has size 6 and which cannot be obtained from local combinatorial data.

From the transport structure we derive an algebraic identity relating the sector–edge incidence matrix to a polynomial in the adjacency matrix, and a corresponding spherical configuration with three distinct inner products.

These results demonstrate how global algebraic and geometric structure can emerge from local transport rules on a combinatorial object.

A Explicit Edge Tables

This appendix records the edge structure of $G_{15} \cong L(\text{Petersen})$ together with the sign function σ defining the signed 2-lift.

We fix an ordering of the 30 edges of G_{15}, as listed in Table **??**. All subsequent references to σ are understood with respect to this ordering.

The value $\sigma(e) \in \{\pm 1\}$ indicates whether the edge e lifts as a parallel edge $(+1)$ or a crossed edge (-1) in the double cover $G_{30} \to G_{15}$.

Edge	Sign	Type
(0,1)	-1	crossed
(0,4)	-1	crossed
(0,5)	$+1$	parallel
(0,8)	$+1$	parallel
(1,2)	-1	crossed
(1,8)	-1	crossed
(1,11)	$+1$	parallel
(2,3)	-1	crossed
(2,11)	-1	crossed
(2,13)	$+1$	parallel
(3,4)	-1	crossed
(3,13)	-1	crossed
(3,14)	$+1$	parallel
(4,5)	-1	crossed
(4,14)	-1	crossed
(5,6)	-1	crossed
(5,12)	$+1$	parallel
(6,7)	-1	crossed
(6,12)	-1	crossed
(6,13)	-1	crossed
(7,8)	-1	crossed
(7,9)	-1	crossed
(7,13)	$+1$	parallel
(8,9)	$+1$	parallel
(9,10)	-1	crossed
(9,14)	-1	crossed
(10,11)	-1	crossed
(10,12)	-1	crossed
(10,14)	$+1$	parallel
(11,12)	$+1$	parallel

B Minimal Cocycle Support

Let

$$S = \{e \in E(G_{15}) : \sigma(e) = -1\}.$$

The set S records the negative edges of the signing defining the signed 2-lift $G_{30} \to G_{15}$.

B.1 Gauge reduction

Two signings related by vertex switching define the same cohomology class. Explicitly, for a function $\tau : V(G_{15}) \to \{\pm 1\}$, the switched signing is given by

$$\sigma'(uv) = \tau(u)\,\sigma(uv)\,\tau(v).$$

Thus the support S is defined only up to switching equivalence.

B.2 Minimal support

We exhaustively search the switching orbit of σ, fixing one vertex to remove the global symmetry.

The computation shows that:

- the support can be reduced to size 6,
- no representative with fewer than 6 negative edges exists,
- there are 40 distinct supports of size 6 within the gauge-fixed search.

This establishes that the minimal weight of the cohomology class is 6.

B.3 Explicit representative

The table below records one representative of minimal support.

Edges in minimal support
$(0, 1)$
$(0, 8)$
$(2, 13)$
$(3, 13)$
$(10, 11)$
$(10, 12)$

B.4 Interpretation

Since the minimal support is nonzero, the cocycle defined by σ is nontrivial in $H^1(G_{15}; \mathbb{Z}_2)$.

The existence of multiple minimal representatives reflects the action of the switching group on the cohomology class.

C Dodecahedron Labeling

We fix a combinatorial model of the dodecahedron used throughout the paper.

C.1 Vertices and edges

Let the dodecahedron have 20 vertices and 30 edges. We label the vertices by a set $V = \{0, 1, \ldots, 19\}$.

Edges are unordered pairs (u, v) with $u, v \in V$. The full edge list is fixed once and for all and determines the combinatorial structure.

C.2 Faces

Each face is a 5-cycle in the edge graph. The 12 faces are recorded as cyclically ordered vertex lists

$$(f_1, f_2, f_3, f_4, f_5).$$

These face cycles determine the local adjacency relations used in the transport construction.

C.3 Flags

A flag is a triple (v, e, f) consisting of a vertex v, an incident edge e, and a face f containing e.

The dodecahedron has 120 flags in total.

C.4 Transport generators

The transport system is defined using local moves on flags:

- vertex move: change the vertex along a fixed edge,
- edge move: change the edge within a fixed face,
- face move: change the face across a fixed edge.

These generate the flag adjacency structure from which the chamber graph G_{60} is constructed.

C.5 Reproducibility

All computations in the paper use a fixed implementation of this labeling, consistent with the edge and face data used in the computational pipeline described in Appendix D.

D Computational Pipeline

This appendix describes the computational procedures used to verify the structural claims in the paper.

D.1 Graph construction

The graph G_{60} is constructed from the 120 flags of the dodecahedron by identifying opposite configurations to obtain 60 chamber states. Adjacency is generated by local transport moves.

The resulting graph is verified to be:

- connected,
- 4-regular,
- with 60 vertices and 120 edges.

D.2 Quotient computation

Successive quotients

$$G_{60} \to G_{30} \to G_{15}$$

are computed by identifying vertices under involutive symmetries.

The graph G_{15} is verified to be isomorphic to $L(\text{Petersen})$ by comparison of adjacency structure.

D.3 Signed lift and cocycle

A signing $\sigma : E(G_{15}) \to \{\pm 1\}$ is computed from the transport rules. The induced double cover is verified to reproduce G_{30}.

Switching equivalence is implemented via vertex sign assignments, and the minimal support is obtained by exhaustive search over the switching orbit.

D.4 Minimal support search

All 2^{14} switching configurations (fixing one vertex) are evaluated. For each configuration, the number of negative edges is recorded.

The minimum support size is found to be 6, and all minimal supports are enumerated.

D.5 Sector construction

The sector–edge incidence matrix M is constructed from the transport data. Matrix products are computed explicitly and verified to satisfy

$$MM^{\mathsf{T}} = A^3 + 2A^2 + 2I,$$

where A is the adjacency matrix of G_{15}.

D.6 Implementation

All computations are carried out using Python scripts operating on explicit adjacency and incidence data. The computations are entirely deterministic and can be reproduced from the data and procedures described above.

E Reproducibility

All results in this paper are based on explicit finite computations on combinatorial data.

E.1 Data

The constructions use:

- a fixed labeling of the dodecahedron (Appendix C),
- an explicit edge list for G_{15} (Appendix A),
- a signing σ defining the cocycle (Appendix A).

E.2 Deterministic procedures

All computations are deterministic and involve:

- graph construction from flag transport,
- quotient identification,
- switching operations on signings,
- exhaustive enumeration over 2^{14} switching configurations,

- matrix computations on integer-valued data.

No randomness or probabilistic methods are used.

E.3 Verification steps

The main verifications can be reproduced by:

- constructing G_{60} from flag data and checking degree and connectivity,
- computing the quotients $G_{60} \to G_{30} \to G_{15}$,
- verifying $G_{15} \cong L(\text{Petersen})$,
- computing the signed lift from σ,
- enumerating switching configurations to confirm minimal support 6,
- computing the matrices A and M and checking

$$MM^{\mathsf{T}} = A^3 + 2A^2 + 2I.$$

E.4 Implementation

All computations are implemented in Python using explicit adjacency lists and incidence structures. The scripts operate directly on the data described in the appendices and require no external libraries beyond standard numerical and combinatorial functionality.

E.5 Scope

The results are fully reproducible from the data and procedures given in this paper. Any implementation following the same definitions will produce identical outputs.

References